

From Physical Part to Digital Precision

I reverse engineer parts of pumps, gearboxes, cardan shafts and structural components. Using a 3D scanner and Geomagic Design X software, I capture real parts and rebuild them as accurate, manufacturable CAD models.

Faro Arm

Geomagic Design X

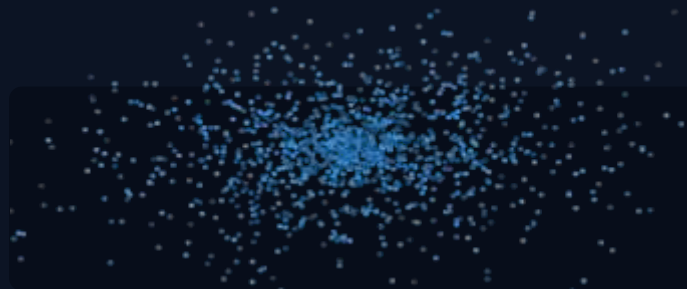
SolidWorks

AutoCAD



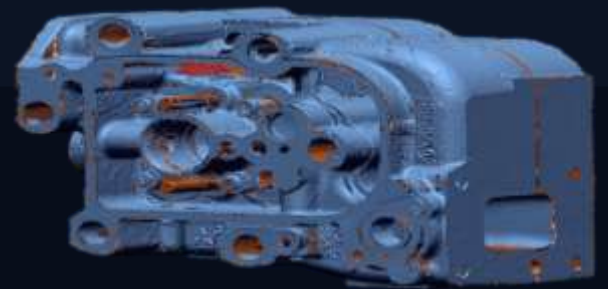
01. GATHERING INFORMATION

Pre-screening, Cleaning of Part, measuring every critical dimension and documenting with sketches.



02. SCANNING & POINT CLOUD

Laser scanning from multiple angles to capture exact surface geometry.

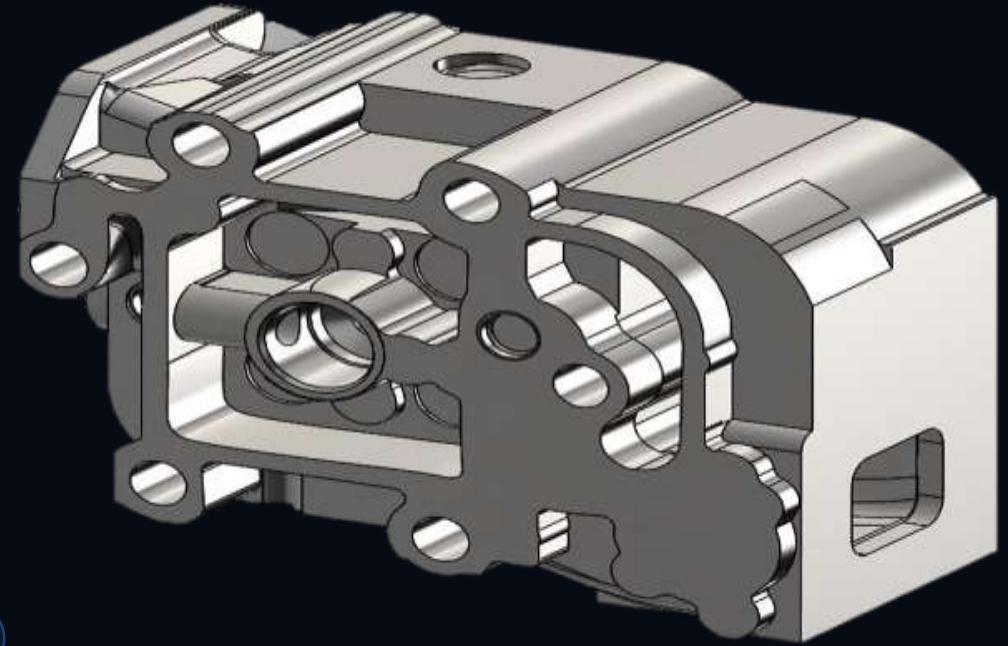


03. MESH MODELLING

Point cloud converted to a polygon mesh with the use of Mesh commands like, Noise Reduction, Region, Alignment, Healing, Decimate etc., ready for CAD integration.

Modelling, Drawing & Manufacturing

From the mesh, I build parametric 3D models with Revolve, Extrude, Sweep & other commands, then release fully dimensioned 2D drawings so the part can be manufactured and verified against its tolerances.

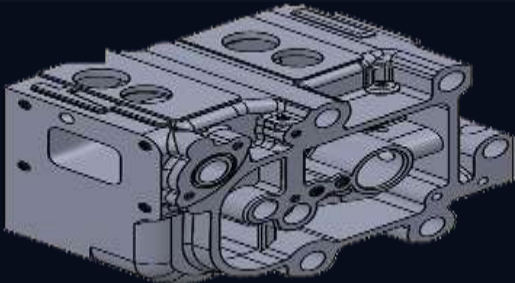


3D Model

2D Drawing

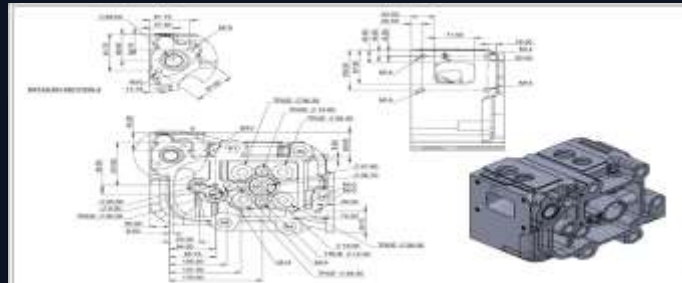
Manufacturing

Quality Control



04. 3D MODELLING

Mesh converted to solid geometry using Different 3D commands like Revolve, Extrude, Sweep etc..



05. 2D DRAWING

Detailed drawings with different Section views, Hidden views, Isometric view, shape, size, BOM, Title box details and specifications for production.



06. MANUFACTURING & QC

Machined to spec, then checked against tolerances and inspected for defects.